

Waypoint-Based Tracking of a Square Trajectory on a Parrot Mambo Minidrone

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Abstract—This report documents trajectory-tracking flights in which a Parrot Mambo minidrone flies a square reference, roughly five to six feet per side, twice in succession over an indoor arena. The flight builds on the MathWorks minidrone waypoint-following example, adapted and tuned by the author, with the square encoded as a sequence of position references. Flying the same lap twice back to back turns the flight into a repeatability test: corner overshoot at the eight sharp heading changes and drift accumulated between laps are directly visible against the arena floor. The deployed flight completes both laps in a single take.

Index Terms—Parrot Mambo, Simulink, waypoint following, trajectory tracking, position control.

I. INTRODUCTION

A hover proves a controller can stand still; a trajectory proves it can go somewhere specific and come back. This flight asks the Mambo to trace a square roughly five to six feet per side, twice, over the arena floor. Repetition is the point: every corner is a step change in commanded direction, and flying the identical lap twice exposes both corner transients and any drift the first lap left behind.

The Parrot Mambo carries the sensor suite the whole series relies on: an ultrasound sensor timing pulses to the floor for altitude, a pressure sensor for altitude changes, an IMU providing linear acceleration and angular rates, and a downward-facing camera whose optical flow contributes horizontal velocity estimates. Four motors in an X configuration are commanded through motor mixing: collective thrust, a differential between the clockwise and counter-clockwise pairs for yaw, and side-to-side speed imbalances for pitch and roll. The airframe is underactuated, with six degrees of freedom and only four inputs, so every lateral motion begins as a deliberate tilt of the thrust vector.

II. SYSTEM ARCHITECTURE

All flight code was developed in Simulink against the Parrot Minidrone Support Package and deployed to the physical airframe, keeping the model structure the support package expects: an estimator consuming the sensor buses, a flight control system, and reference logic that the author modified per task. Structuring the work as small reusable subsystems made iteration and debugging on hardware considerably less painful.

The model derives from the minidrone waypoint-follower example project, whose structure the author adapted: the square's corners become waypoints in the reference logic, and the flight control system tracks the resulting position reference through the cascaded position and attitude loops.

III. METHOD

The square is encoded as a timed sequence of position references, flown at constant altitude, with two full laps commanded back to back. Between corners the position loop drives the drone along each edge; at corners the reference direction changes sharply, exercising the controller's transient response eight times across the flight.

IV. FLIGHT TESTING AND RESULTS

The deployed flight completes two consecutive laps of the square in one take over the checkerboard arena. Flying the same pattern twice in succession makes lap-to-lap consistency directly visible, and the second lap tracks the same ground region as the first without a reset between laps.

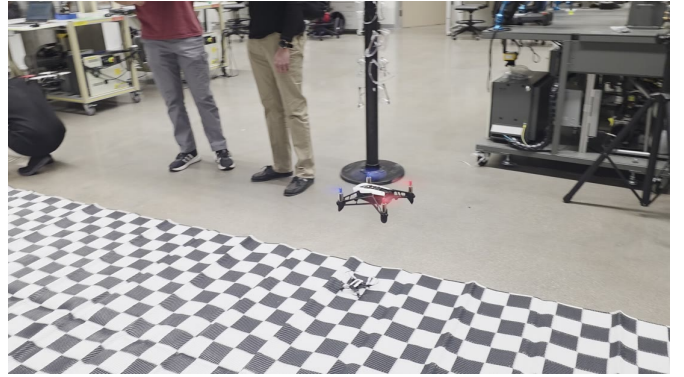


Fig. 1. The Mambo over the checkerboard arena during the square-trajectory flight.

V. CHALLENGES

- Corner transients: four step changes in commanded direction per lap, eight across the flight.
- Drift accumulating across laps with no external positioning reference indoors.
- Separating controller error from estimation drift when judging the flown path by eye.

VI. CONCLUSION

The flight demonstrates repeatable position tracking: the same square, flown twice in one take on real hardware, validated the position controller for the vision-guided missions that follow in the series.

APPENDIX: MATLAB FUNCTION CODE

The following MATLAB Function blocks are the author's custom code from the deployed Simulink model, reproduced verbatim.

Listing 1. Earth-to-body frame error transformation (flight control model)

```
function xy_err_body = fcn(yaw,xy_err_earth)
transformation = [cos(yaw), sin(yaw); -sin(yaw), cos(yaw)];
```

```
xy_err_body = transformation * xy_err_earth;
```

```
end
```

ACKNOWLEDGMENT

The author completed this work as part of RAS 546, Robotic Systems II, in the MS Robotics and Autonomous Systems program at Arizona State University, and thanks Prof. Sangram Redkar and the course teaching assistants for guidance on the Parrot Mambo platform, flight-arena logistics, and safe flight operations.

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